

CRMP Project Proposal

Course Code:

CSCC01H3

Team Name:

Make No Mistakes

Team Members + Contributions (For Attendance Purposes):

- Xin Chen: Personas
- Andy Yang: Use Case Model
- Le Xu: Use Case Scenarios
- Vincent Lam: User Interface Sketches (Analyze Submissions)
- Amelie Breton: User Interface Sketches (Counter Proposal, View Proposals), Use Case Model

Personas



Jessica Chen

Age: 34 **Gender:** Female

Location: Markham, Ontario

Occupation: High-school math teacher

Jessica has recently become interested in federal electoral redistribution because a proposed boundary change may separate her neighbourhood from the rest of her community. She wants to understand what effect the proposed boundary change really will make in an easy way and submit her feedback on that.

Background

Motivation

- Jessica believes community voices should be included in political decision-making.

Goals

- Understand how the proposed boundary change will affect her neighbourhood
- Submit comments or feedback to the government easily so that she can participate in the democratic consultation process
- Compare current and proposed electoral districts

Frustrations

- Government redistribution maps are hard to understand
- No existing platforms to share concerns with the community

Tech Knowledges

- Comfortable using websites and mobile apps.
- No advanced experience in GIS



Tom Davidson

Age: 50

Gender: Male

Location: Downtown, Toronto

Occupation: Provincial electoral boundary commission

Tom has years of experience contributing to public administration and policy analysis. During redistribution consultations, he is responsible for reviewing public submissions and evaluating whether proposed changes meet the needs of the population and legal requirements.

Background

Motivation

- Tom wants a system that improves efficiency while maintaining transparency in the redistribution process

Goals

- Efficiently review and categorize public submissions
- Export organized reports for the commission's final review

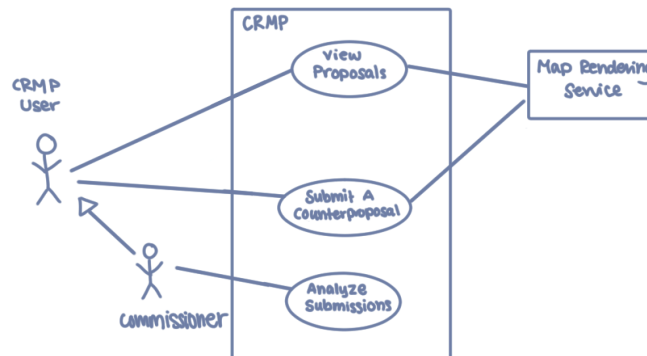
Frustrations

- Current consultation systems rely heavily on email and manual review
- Large volumes of unstructured comments are difficult to organize

Tech Knowledges

- Experienced with administrative software, spreadsheets, and government data systems, but not a professional programmer.

Use Case Model



View/Browse Proposals

(Category: Public Browsing)

Users may browse through proposals on a public account, then view related user submissions. A public user opens the Canadian Redistribution Map Platform (CRMP) to explore proposed federal elector-district boundaries. The user may filter and sort through the browsing page to quickly view submissions relevant to them. The user can pan and zoom the interactive map, compare proposed boundaries with existing legal boundaries, and click on a riding to view important statistics such as population, deviation from the electoral quotient, and communities of interest. The user may also read through the user submissions left by others. This use case supports informed public participation in the redistribution process.

Submit A Counter-Proposal

(Category: Public Submission)

After logging in to the CRMP, the user has the option to draw a counter proposal. They may draw an alternative configuration for boundaries, given that the form of the submission is valid (for example, the submission must not have any disconnected pieces) and the population distribution would be updated in real time for the affected ridings to help the user and commissioners check the validity of the mapping. The system then timestamps the submission, associates it with the user's account, and assigns a unique public reference number. The user can later review the submission status through the "My Submissions" page. This helps admins gain more thorough feedback on what the public wants.

Analyze Submissions for Public Concern

(Category: Commissioner Dashboard)

A commissioner logs into the CRMP administrative dashboard to review submissions received during the consultation process. The commissioner can filter submissions by riding, date, or submission type, apply organizational tags such as "support" or "technical correction," attach internal notes, and export selected submissions for inclusion in official reports. The dashboard also provides real-time consultation statistics and geographic summaries to support efficient decision-making. A heatmap is also visible so the commissioner can determine which areas are of the greatest public concern.

Use Case Scenario Models

Browse Proposals and Provide Feedback

Trigger: User enters site homepage

Preconditions: User is either not logged into any account, or logged into a public user account

Interested stakeholders: public users

Active stakeholders: public users

User	Application	Database
1. User enters site homepage.		
	2. Application requests data of the commission's proposed boundary map, and riding-level statistics.	
		3. Database returns the requested data to the application.
	4. Application displays requested data to the user in the appropriate page views of the requested data; displays a summary panel with riding-level statistics, when a particular riding is selected.	
5. User browses the map as they wish, with access to features that are permitted relative to their account status (not logged in; public user).		
Alternative 1 (From step 4)		
4-1. User toggles second layer of the map to show currently existing boundaries.		
	4-2. Application requests the current information of the existing boundary map from database.	
		4-3. Database returns the requested information to the application.
	4-4. Go to Normal Case step 4.	
Alternative 2 (From step 4)		

4-1. User wants to submit one of the following: written comment, boundary objection, counter-proposal.		
	4-2. Application checks for user account status (not logged in; public user account).	
	4-3. If user is logged in, go to step 4-6.	
	4-4. If user is not logged in, the application displays a create account view to user.	
4-5. User creates an account and logs in.		
	4-6. User is redirected to view for creating submissions.	
4-7. User completes the corresponding form. For Counter Proposal submissions, refer to the scenario 'Submit a Counter Proposal'.		
	4-8. Application uploads the submission with identifying information (timestamp, submitter info, public reference number), to the database.	
		4-9. Database records the submission for review by the Commission.
	4-10. User is notified of the recorded submission, and is redirected to the homepage.	
4-11. Go to Normal Case step 1.		
Exception (From step 2)		
	2-1. The application failed to request the required information from the database. (or couldn't receive the information)	
	2-2. Inform the user that an error has occurred and prompt them to reload the page. (provide an useful error code if possible)	
2-3. User acknowledges the error and reloads the page.		
2-4. Go to Normal Case step 1.		

Submit A Counter-Proposal

Trigger: User enters counter proposal submission page

Preconditions: User is logged into a public user account

Interested Stakeholders: public users, commissioners

Active Stakeholders: public users

User	Application	Database
1. User draws alternative boundary configuration		
	2. Application confirms validity of the configuration, parses it, and reports back relevant statistics in real time.	
3. User confirms submission of counter proposal		
	4. Application uploads submission with identifying information (timestamp, submitter info, public reference number) to the database.	
		5. Database saves the submission for review by the commissioner later.
	6. Application redirects the user to the main page and informs them of the saved submission.	
Alternative 1 (From step 2)		
	2-1. Application detects invalid configuration of counter proposal.	
	2-2. Application blocks the submission of current counter proposal and informs user with a prompt.	
2-3. User acknowledges the prompt and reconfigures the counter proposal.		
	2-4. Go to Normal Case step 2.	
Alternative 2 (From step 2)		
2-1. User wants to exit the counter proposal submission page.		
	2-2. Application displays a prompt to confirm user intent.	

2-3. If user confirms, redirect user to the page they wished to go to. (End of scenario)		
2-4. If user wishes to stay on the counter proposal submission page, remove the confirmation prompt.		
	2-5. Go to Normal Case step 2.	
Exception (From step 4)		
	4-1. Application fails to upload the submission.	
	4-2. Application displays an error message informing the user of failure to save the submission.	
4-3. User acknowledges the error message.		
	4-4. Application displays the user's unchanged counter proposal configuration for resubmission.	
	4-5. Go to Normal Case step 2.	

Analyze Submissions for Public Concern

Trigger: User enters commissioner dashboard page

Preconditions: User is logged into an invitation-only account for commissioners

Interested stakeholders: commissioners, public users

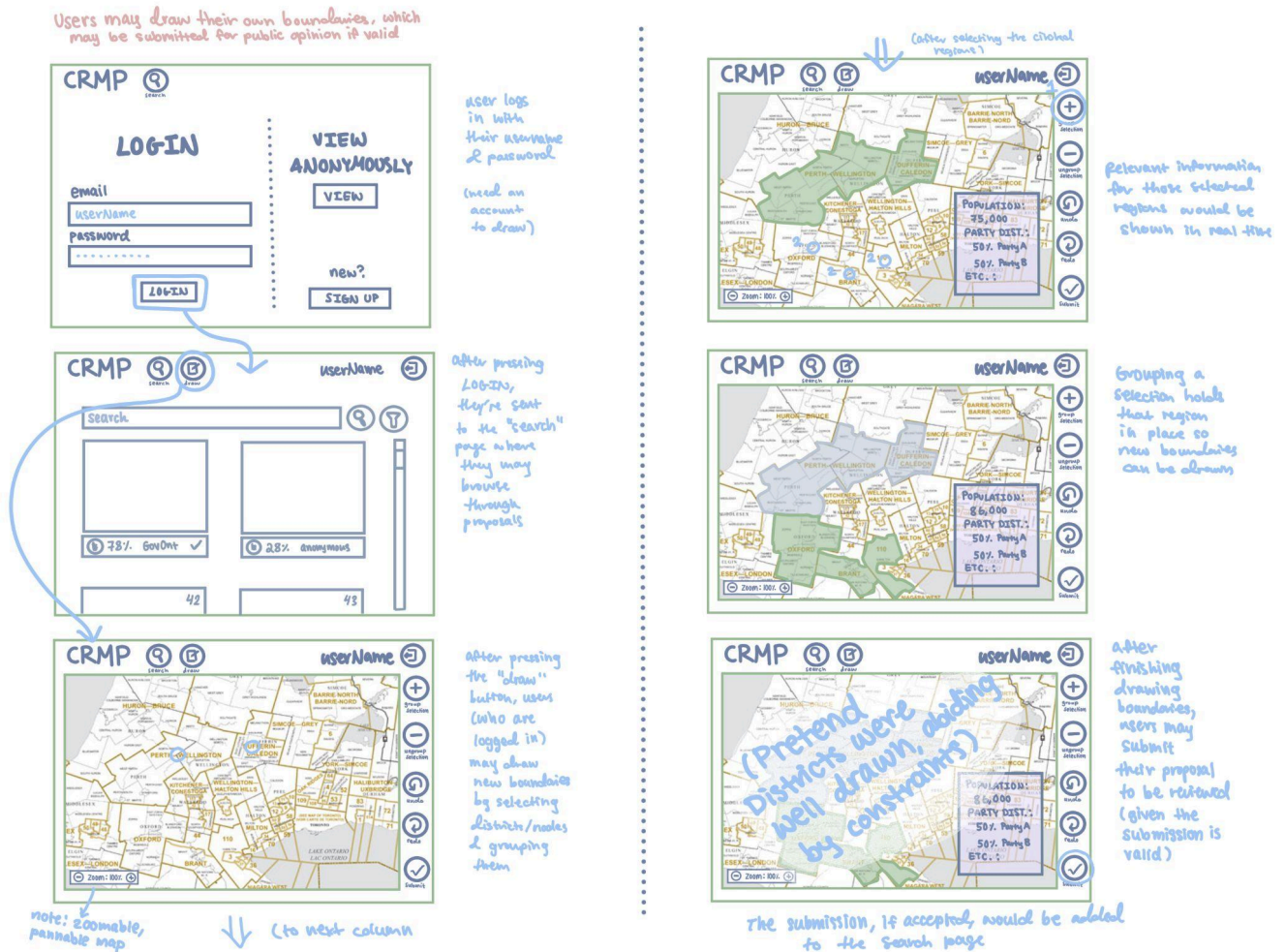
Active stakeholders: commissioners

User	Application	Database
	1. Application requests aggregate information from database (non-exhaustive) in real time: 1. Submission volume over time. 2. Breakdown of user submissions (by type, etc.). 3. Geographic heatmap, for ease of identifying ridings of greatest public concern. 4. For each riding, a panel that displays information of the currently selected riding such as: submission count, ratio of supporting/opposing submissions, most frequently recurring themes.	
		2. Database returns the requested information to application.
	3. Application displays the requested real time information as an overview.	
4. User requests for exporting filtered subsets of submissions with their associated tags to CSV/PDF files.		
	5. Application generates the requested file for user to download.	
Alternative 1 (From step 3)		
3-1. User selects a riding.		
	3-2. Application displays more information for that specific riding. (riding summary view)	
3-3. User processes items from the submission queue sequentially. (with keyboard-driven tagging)		
	3-4. Application updates the view as user processes each item.	

3-5. User decides to stop processing items and return to riding summary view.		
	3-6. Application redirects user to riding summary view.	
3-7. User chooses to return to dashboard view.		
	3-8. Application redirects user to dashboard view.	
	3-9. Go to Normal Case step 1.	
Alternative 2 (From step 3)		
3-1. User goes into submission viewing page. (Supports searching, sorting, filtering)		
	3-2. Application displays submissions as requested.	
3-3. User selects a particular submission to view.		
	3-4. Application displays information (submission details, submitter, time, tags, etc.) of selected submission, also offers individual submission exporting (CSV/PDF).	
3-5. Go to step 3-1 if user continues to browse individual submissions.		
3-6. User returns to dashboard.		
	3-7. Go to Normal Case step 1.	
Exception (From step 2)		
	2-1. Application fails to receive information to load dashboard properly. (Couldn't send the request, or possibly failed to receive information)	
	2-2. Application informs the user that an error has occurred and prompts the user to reload the page and try logging in again. (if possible, provide an useful error code)	
2-3. User reloads the page.		
	2-4. Go to Normal Case step 1.	

User Interface Sketches

Scenario: Submit a Counter-Proposal



Users that are logged in have the ability to create counter proposals, given that their proposal abides by the necessary conditions to be valid.

Scenario: Browse Proposals

users may browse through proposals anonymously, though functionality is limited

CRMP

OR (clicks lead to search anonymously if not signed in)

LOGIN

Email

Password

LOGIN

VIEW ANONYMOUSLY

VIEW

new? **SIGN UP**

Users can view anonymously by either clicking on "search" or "view" in the log in page.

CRMP

LOGIN?

Search

(preview of map goes here) 78% #117

(preview goes here) 28% #118

From there, the user is sent to the search page, where they may look through proposals.

CRMP

LOGIN?

Search

FILTERS

Sort By

- ☒ popularity
- ☐ approval rating
- ☐ date posted

Include Only

- ☐ criteria 1
- ☒ criteria 2
- ☐ ETC.

CLEAR **APPLY**

Users may then select filters of search for a particular issue.

CRMP

LOGIN?

Search

(map) 78% #47

(map) 96% #11

By clicking on a thumbnail photo, users can get the expanded view for a proposal.

CRMP

LOGIN?

#47 98%

COMMENTS

NAME 123

View suggestion

I think these districts should be bigger, since the population is smaller up there!

ANOTHER USER 456

View suggestion

THIRD USER 789

I think this is lovely! :)

2. zoom -

As the user is anonymous, they are unable to vote or comment (subject to change).

However, they may look at current ratings, read comments, & view suggestions.

(If logged in, the user would be able to make a comment, or if they agree, create a suggestion with the option to test for an early description of their issue)

CRMP

LOGIN?

#47 98%

COMMENTS

NAME 123

View suggestion

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THIRD USER 789

I think this is lovely! :)

2. zoom -

Clicking "view suggestion" expands the comment's suggestion into the map, which can easily be cleared to return to the original map.

Even anonymous users are able to sort and filter through proposals (browsing), and determine through user submitted comments on each proposal what the public perception is on proposals and counter-proposals.

Scenario: Analyze Submissions for Public Concern

The sequence of screenshots illustrates the navigation path for analyzing submissions:

- Summary View:** The initial screen shows a summary of submissions for a specific riding, including a bar chart for submission volume, a pie chart for submission types (Comments, Objections, Proposals), and a list of available tags.
- Heatmap View:** The user clicks on the 'Heatmap' tab, displaying a map of Ontario with a color-coded heatmap representing submission volume across different ridings.
- Province Selection:** The user clicks on the 'Province' dropdown menu, which shows a list of provinces and territories.
- Submission Details:** The user selects a specific riding, leading to a detailed view of submissions for that riding. This view includes a search bar, a list of submissions with filters (Comment, Objection, Proposal), and a sidebar with internal notes and tags.

(Can be accessed from any page)

Commissioners have the ability to view a heatmap of comments and objections per riding, and view summary statistics for submissions (which they may then download as a PDF or CSV)